MACROECONOMICS

MACRO-54

Time: 6 minutes

Outcome: Calculate the real exchange rate and use PPP as a long-run benchmark.

Difficulty: Intermediate

Real Exchange Rates and Purchasing Power

THE CORE IDEA

Let e be foreign currency per domestic currency, P the domestic price level, and P* the foreign price level. The course convention is epsilon = e x P / P*. A higher epsilon is a real appreciation: domestic goods are relatively more expensive. Purchasing-power parity suggests epsilon tends toward 1 when comparable traded goods can be arbitraged, but transport costs, trade barriers, nontraded goods, and market frictions prevent exact short-run equality.

HOW TO RECOGNIZE IT

- Nominal e compares currencies; real epsilon compares basket prices after conversion.
- A higher e or P raises epsilon; a higher P* lowers epsilon, other things equal.
- Real appreciation tends to reduce exports and raise imports.
- Relative PPP links currency changes to inflation differentials over longer horizons.

WATCH OUT

Do not invert the formula by importing another textbook's quotation. Write e's units first. Here e is foreign currency per domestic currency, so epsilon = eP/P*.

WORKED EXAMPLE: CONVERTING TWO BASKETS

KEY RELATIONSHIPS

- e: foreign currency / domestic currency
- epsilon = e x P / P*
- Higher epsilon = real appreciation
- PPP benchmark: epsilon near 1
- Approx.: % change epsilon = %e + domestic inflation - foreign inflation

A U.S. basket costs $50, a comparable foreign basket costs 40 foreign units, and e = 0.80 foreign units per dollar. Then epsilon = 0.80 x 50 / 40 = 1.00. If the U.S. price rises to $55 with e and P* fixed, epsilon becomes 1.10: a 10% real appreciation.

CHECK YOURSELF

With e and P fixed, foreign prices rise. Does epsilon rise or fall under the course formula?

READY? Return to the game and master this concept.